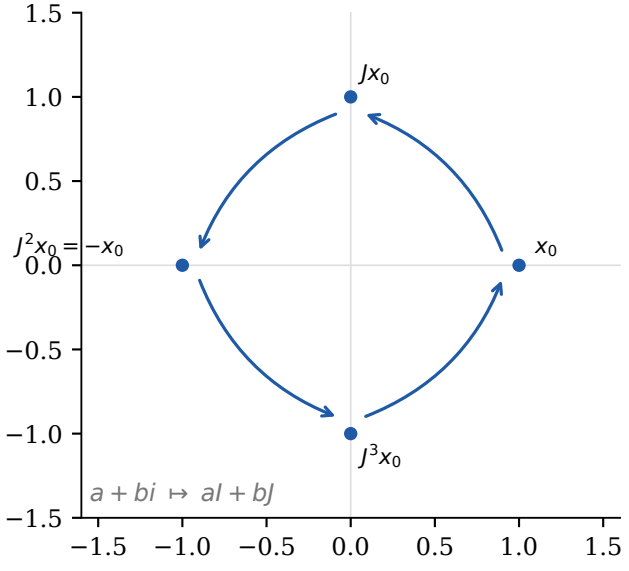
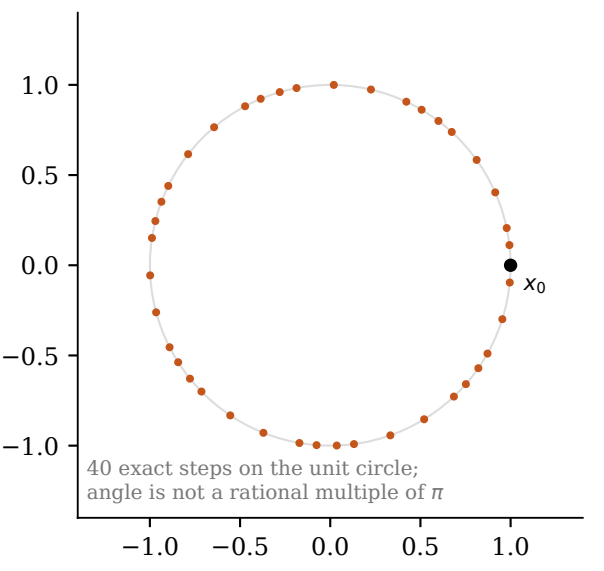

(a) $i \mapsto J$, $J^2 = -I$: four quarter-turns
Jx_0
$J^2x_0 = -x_0$
x_0
J^3x_0
$a + bi \;\mapsto\; aI + bJ$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


(b) $(3 + 4i)/5$: a rotation that never returns
x_0
40 exact steps on the unit circle;
angle is not a rational multiple of π
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0